

Introduction to Biophysics II: Quantum Biology

Phenomena:

- Photosynthesis
- Vision
- Avian compass
- Sense of smell

Methods:

- Optical spectroscopy
- Modeling complex systems

Theory (Quantum mechanics):

- Absorption/emission of light
- Electron transfer/tunneling
- Molecular vibrations
- Coupled electronic states (excitons)
- Forster and Dexter energy transfer
- Redfield theory of energy transfer

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Introduction to Biophysics II: Quantum Biology

Lecture 01

Introduction

Photosynthesis

Electron wavefunctions

- Syllabus
- What is life?
- Photosynthesis: an overview
- Electron wavefunctions (hydrogen atom)

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About the course

Meeting Time and Place:

TTh 4:30-5:45

PHYS 338

Instructor:

Prof. Sergei Savikhin

Office: PHYS 64

email: sergei@purdue.edu

Office hours: by request

Brightspace: <https://purdue.brightspace.com/d21/le/content/1499248/Home>

Course credits: 3

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About the course

Recommended Material Related to the Course:

https://web.ics.purdue.edu/~sergei/classes/quantum_biology

Recommended Text(s):

- Modern Optical Spectroscopy. With exercises and examples from biophysics and biochemistry. W.W. Parson. 2nd edition, Springer 2015
- Quantum Effects in Biology. Eds. M. Mohseni, Y. Omar, G.S. Engel, M.B. Plenio. Cambridge 2014

Suggested additional readings:

- Philip Nelson, Biological Physics, 2008: chapters on cell and amino acid structure
- Glaser "Biophysics"
- Schrödinger "What is life"
- Meyer B. Jackson, Molecular and Cellular Biophysics, 2006.
- Introductory biochemistry text, for example, Jeremy M. Berg, John L. Tymoczko, and Lubert Stryer "Biochemistry".
- Selected chapters from Eric R. Kandel and James H. Schwartz, Principles of Neural Science, 2000

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About the course

Grading:

Homeworks/ projects	50%
Oral presentation of a paper from a research journal. (Select biophysics-oriented paper from recent publications)	50%

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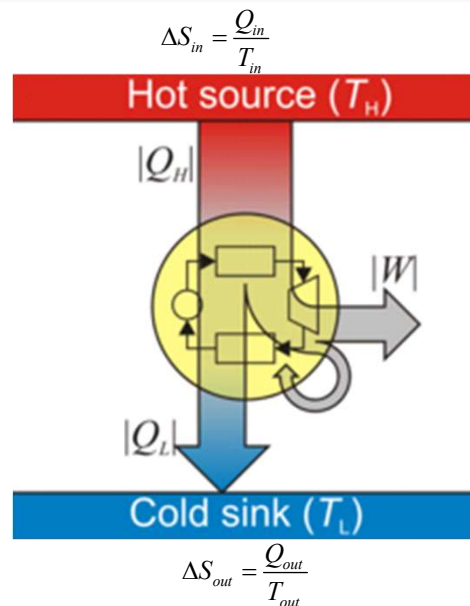
What is life?



<https://www.northcoastjournal.com/life-outdoors/fighting-entropy-11177235>

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What is life?... Heat engine...



$$\Delta S_{out} \geq \Delta S_{in}$$

$$\frac{Q_{out}}{T_{out}} \geq \frac{Q_{in}}{T_{in}}$$

$$\begin{cases} \frac{Q_{out}}{T_{out}} \geq \frac{Q_{in}}{T_{in}} \\ W = Q_{in} - Q_{out} \end{cases}$$

$$W \leq Q_{in} - \frac{T_{out}}{T_{in}} Q_{in}$$

$$W \leq \frac{T_{in} - T_{out}}{T_{in}} Q_{in}$$

$$\eta \leq \frac{T_{in} - T_{out}}{T_{in}}$$

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Life: lowering entropy

Photon at 6000 K: ~ 2 eV

$$\eta \leq \frac{T_{in} - T_{out}}{T_{in}}$$

Photon at 300 K: ~ 0.025 eV



Net entropy increases – photosynthesis is a middleman, decreases entropy locally

ADP → ATP: 0.31 eV

Synthesis of a typical protein (200 amino acids): ~250 eV

<https://www.northcoastjournal.com/life-outdoors/fighting-entropy-1117235>

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Lectures 01-05

Photosynthesis:

Overview

Physical principles governing photosynthesis

Electronic properties of complex molecules

Absorption of light

Energy transfer

Electron transfer

Reading:

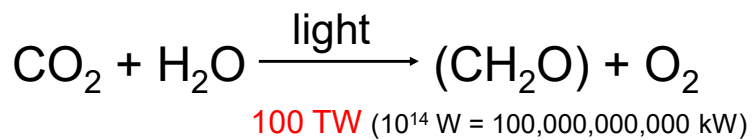
1. WW Parson. Modern Optical Spectroscopy. Chapter 2; Chapter 1.
2. DJ Griffiths. Introduction to Quantum Mechanics. Chapter 9, Time-dependent Perturbation Theory

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Photosynthesis



Photo – light
Synthesis - building



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Top 10: Life's greatest inventions

1. Multicellularity
2. The eye
3. The brain
4. Language
- 5. Photosynthesis**
6. Sex
7. Death
8. Parasitism
9. Superorganism
10. Symbiosis



04 September 2006
NewScientist.com

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Photosynthesis: energy

Photosynthesis: ~100 TW

Human society in 2012 : ~17 TW
in 2023: ~ 18 TW (electrical ~3.2 TW)

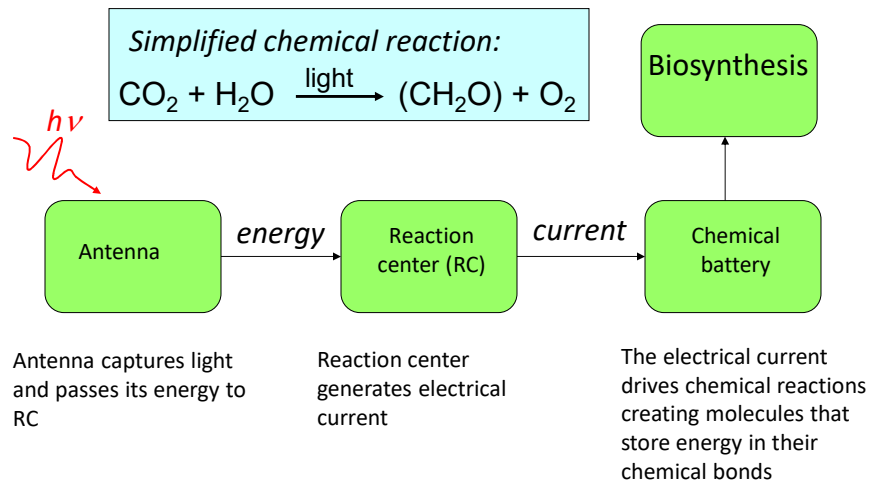
Sunlight total : 120,000 TW (fall on Earth)
usable : 800 TW (can harvest)

Note: most of the new energy comes from renewables

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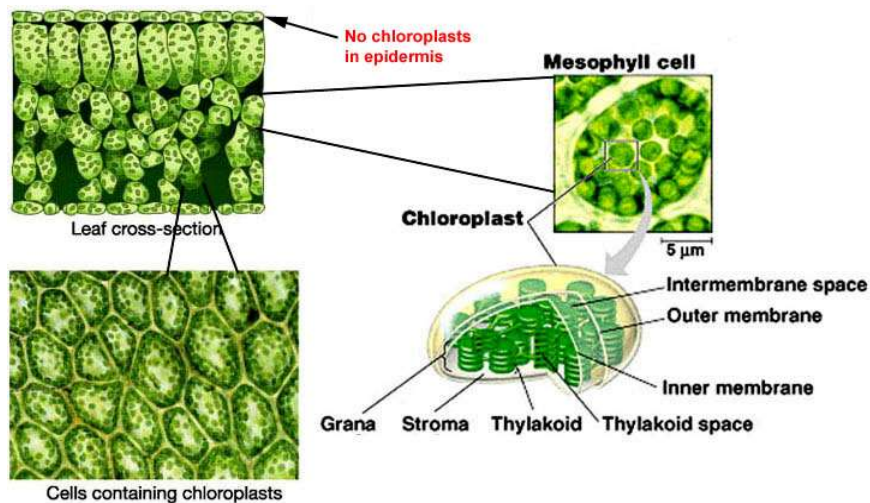
Photosynthesis

Simplified block-diagram:



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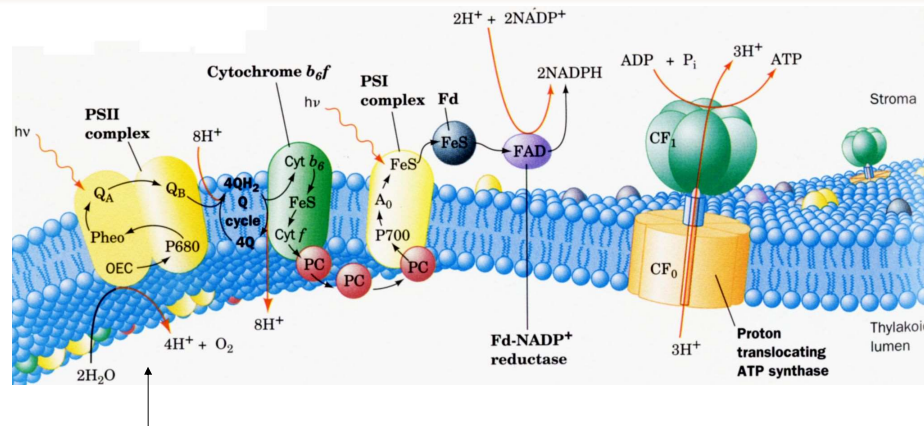
Photosynthetic apparatus of plants



Photosynthetic powerplant is located in thylakoid membranes

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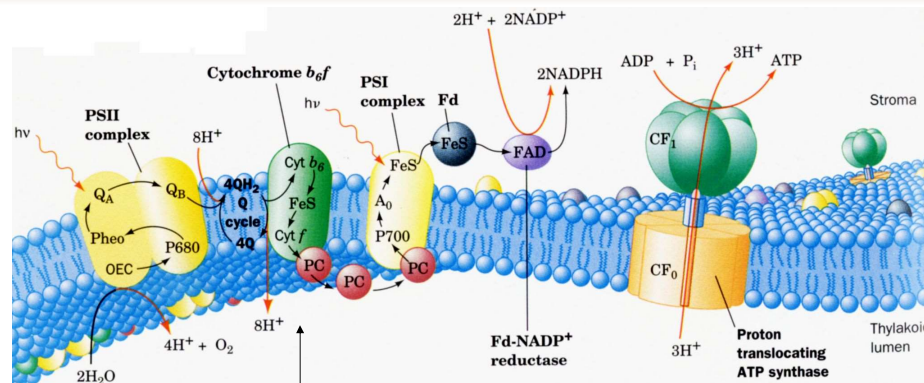
Photosynthetic apparatus: PS II



Photosystem II (PS II) complex uses energy of light to split water and generate electron and proton current.
Oxygen molecules are byproduct of this processes
The PS II is extremely strong oxidant - it can oxidize water!

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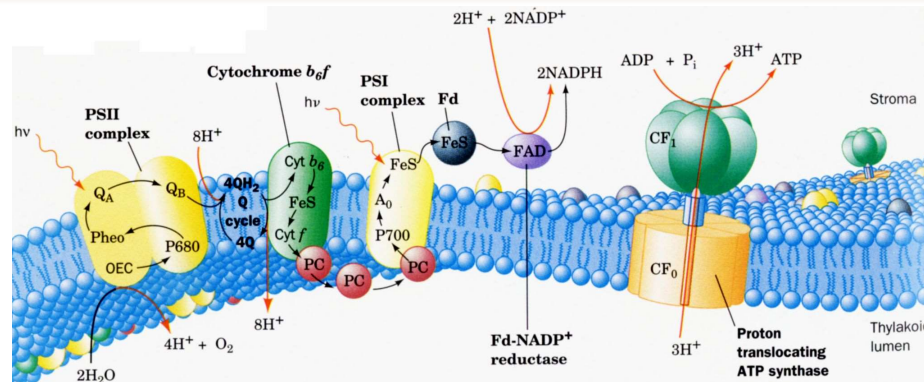
Photosynthetic apparatus: cytochrome b_6f



Cytochrome b_6f complex transfers electrons (current) from PS II to photosystem I (PS I) complex.
In addition, the fraction of this electron current is used to transports protons across the thylakoid membrane.
Plastoquinones (Q , QH_2) and plastocyanins (PC) serve as intermediate electron carriers

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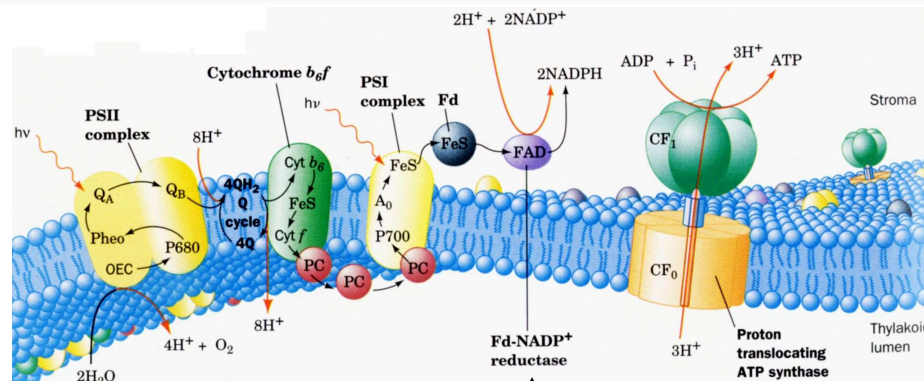
Photosynthetic apparatus: PS I



Photosystem I gets electrons from PS II via cytochrome b_6f and uses light to boost the energy of these electrons and transfer them to mobile ferredoxin (Fd) complex

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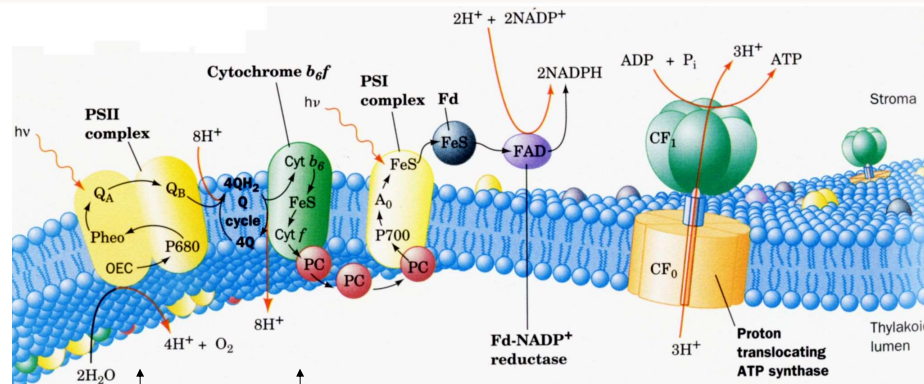
Photosynthetic apparatus: FAD



Ferredoxin (Fd) uses the energy of the electron to reduce NADP^+ molecule to NADPH and store the energy in chemical bond of NADPH. This reaction is mediated by ferredoxin-NADP⁺ reductase (FAD)

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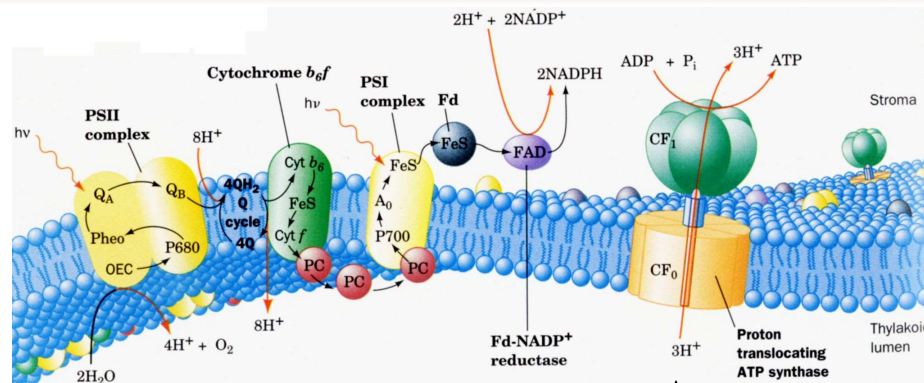
Photosynthetic apparatus: proton pumps



PS II and cytochrome b_6f also pump protons creating electrochemical potential across the membrane (like charging a capacitor). That creates strong electric field across the membrane.

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Photosynthetic apparatus: ATP synthase

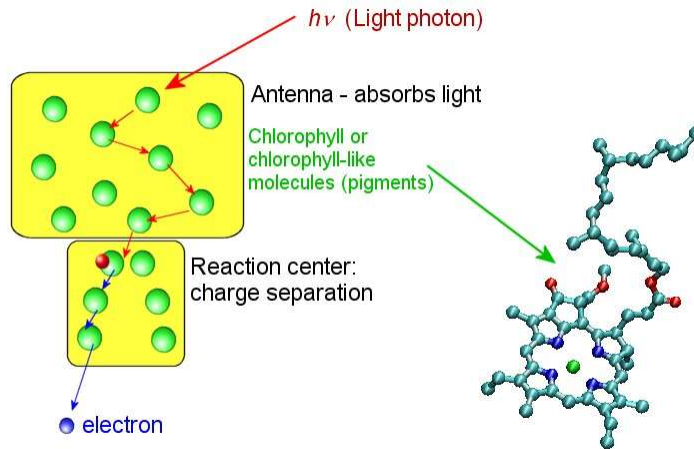


This strong electric field drives protons through a molecular size 'turbine' motor called ATP synthase. The axle of that molecule spins and mechanical force drives chemical reaction transforming ADP molecule to ATP - the energy is stored in extra chemical bond

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Antenna and reaction center

- Large variety of different photosynthetic systems (plants, bacteria, algae) share two common elements: light-harvesting **antenna** and **reaction center** (RC).



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Questions

- How is light energy stored by a molecule?
- How is that energy transferred toward the reaction center?
- How is the energy used to initiate charge separation?
- How can electron be transferred through a protein?

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Wave function

Particles (electrons) are represented by a wave function:

$$\int_a^b |\Psi(x,t)|^2 dx = \left\{ \begin{array}{l} \text{Probability of finding the particle} \\ \text{between } a \text{ and } b, \text{ at time } t. \end{array} \right\}$$

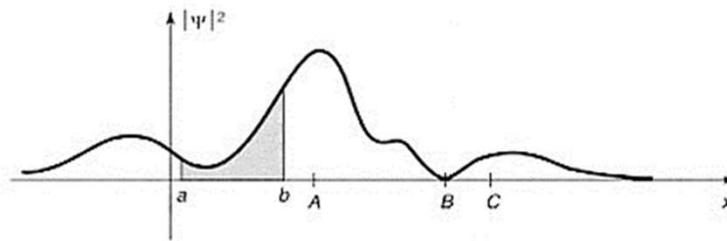


FIGURE 1.2: A typical wave function. The shaded area represents the probability of finding the particle between a and b . The particle would be relatively likely to be found near A , and unlikely to be found near B .

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Schrödinger Equation

Evolution of a particle is described by Schrödinger Equation (1926)
(it is a wave equation):

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = \hat{H}\Psi$$

(in one dimension)

$\Psi(x,t)$ - Wave function

m - mass of the particle

$V(x,t)$ - potential energy

$$i \equiv \sqrt{-1}$$

$$\hbar \equiv h/2\pi = 1.054572 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V$$



Erwin Rudolf Josef
Alexander Schrödinger
1887 - 1961

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Operators

Physical quantities are represented by operators:

$$\text{Expectation value: } \langle A \rangle = \langle \Psi | \hat{A} | \Psi \rangle \equiv \int \Psi^* \hat{A} \Psi dr$$

Examples:

$$\langle x \rangle = \int \Psi^* \cdot (x) \cdot \Psi dx$$

$$\langle p \rangle = \int \Psi^* \cdot \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \cdot \Psi dx$$

$$\langle E \rangle = \int \Psi^* \hat{H} \Psi dx$$

Operators:

$$(x) \quad - \text{ position}$$

$$\hat{p}_x = \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) - \text{ momentum}$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \quad \text{energy}$$

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Hermitian operators

- *Measurable* physical quantities must be represented by *Hermitian* operators! – since measurable quantities are *real* numbers:

$$\langle \Psi | \hat{A} | \Psi \rangle = \langle \Psi | \hat{A} | \Psi \rangle^*$$

Or:

$$\langle \Psi_a | \hat{A} | \Psi_b \rangle = \langle \Psi_b | \hat{A} | \Psi_a \rangle^*$$

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Time-independent SE

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \quad i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi$$

- *Time-independent* Schrödinger Equation:
when H depends on coordinates only and not on time:

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r})$$

$$\Psi(\vec{r}, t) = \psi(\vec{r}) \exp\left(-\frac{2\pi i E}{\hbar} t\right)$$

Solutions to time-independent SE are *stationary* states with defined energy E

Any particle state can be represented as a superposition of stationary states, each oscillating at frequency defined by energy E .

Solutions are orthogonal and normalizable:

$$\langle \psi_i(\vec{r}) | \psi_k(\vec{r}) \rangle = \delta_{ik}$$

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Free vs confined particle

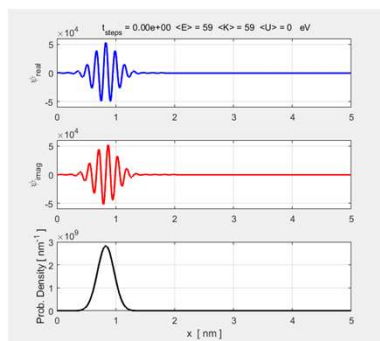
Quantum mechanics:

- *Free particle:*

Wave, Continuous spectrum of E

$$\psi(x) = A \exp\left(i\sqrt{2mEx}/\hbar\right)$$

Superposition of multiple ψ

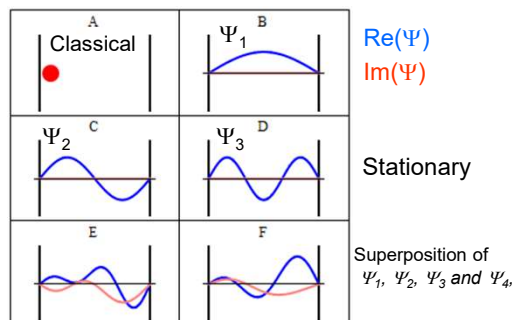


Particle in a box:

Confined wave, discrete E

$$\psi_n(x) = (2/l)^{1/2} \sin(n\pi x/l)$$

$$E_n = n^2 \hbar^2 / 8ml^2$$



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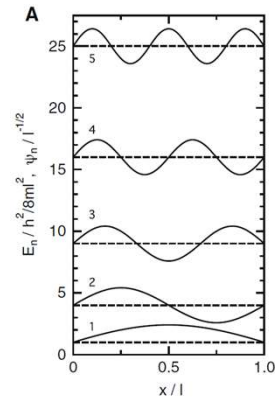
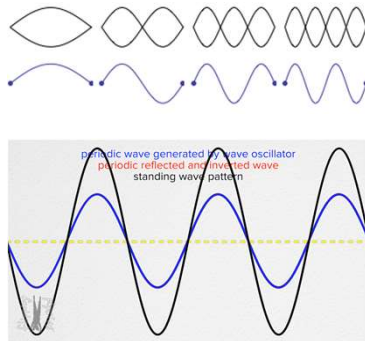
Free vs confined particle

Particle in a box:
Confined wave, discrete E

$$\psi_n(x) = (2/l)^{1/2} \sin(n\pi x/l)$$

$$E_n = n^2 h^2 / 8ml^2$$

Standing waves:



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Particle in harmonic trap

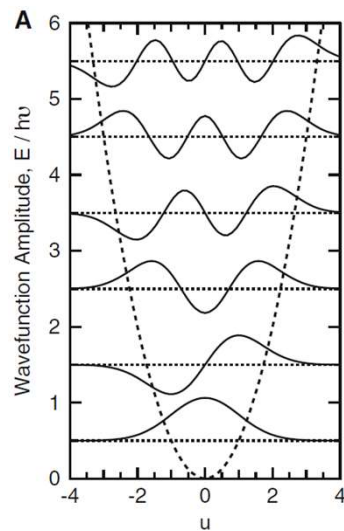
Harmonic potential: $V(x) = \frac{1}{2} m \omega^2 x^2$

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega$$

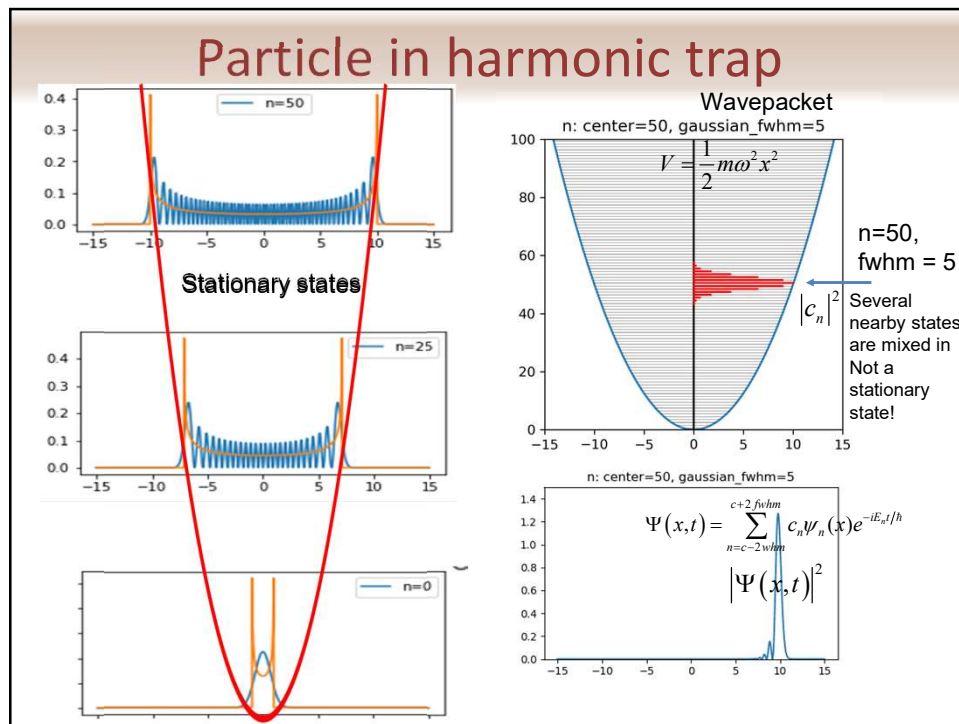
$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}$$

Hermite polynomial

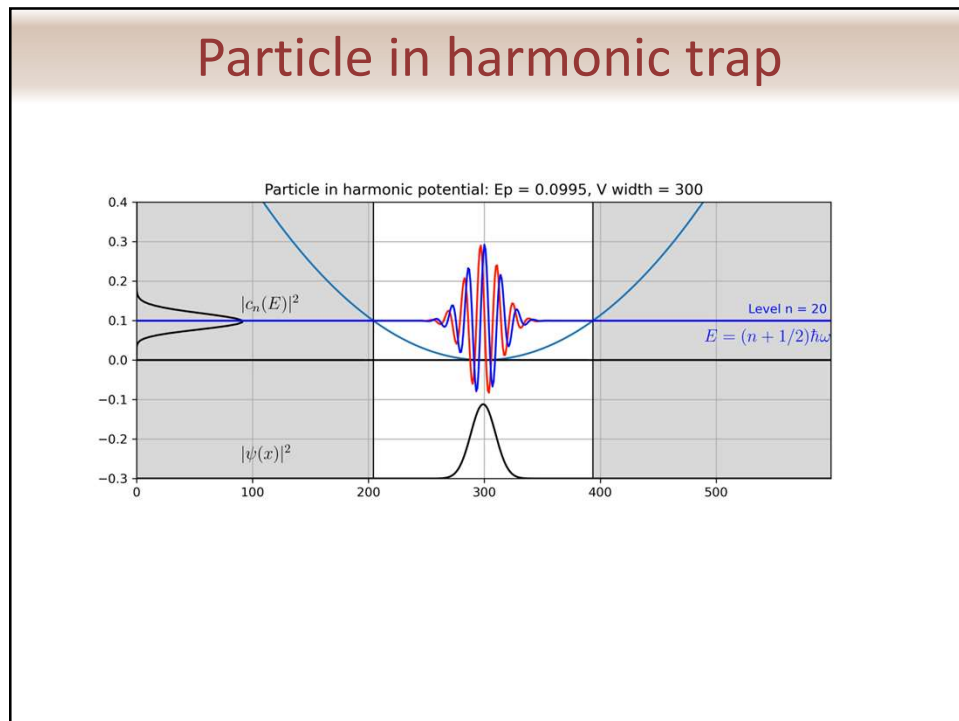
$$\xi \equiv \sqrt{\frac{m\omega}{\hbar}} x$$



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Atomic orbitals

Atomic orbitals: solution for 3D Schrödinger equation for Hydrogen-like atom with 1 electron

$$\psi_{nlm} = R_{n,l}(r)Y_{l,m}(\theta, \phi) \quad (\text{in spherical coordinates})$$

$$E_n = -16\pi^2 Z^2 m_r e^4 / n^2 h^2 \quad Z - \text{nuclear charge}$$

Quantum numbers:

$n = 1, 2, 3, \dots$ ← Principal quantum number: (defines energy)

$l = 0, 1, 2, \dots, n-1$ Azimuthal quantum number (angular momentum)

$m = -l, \dots, 0, \dots, +l$ Magnetic quantum number

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Atomic orbitals

Atomic orbitals for hydrogen-like atom of charge Z

Table 2.1 Hydrogen-atom wavefunctions

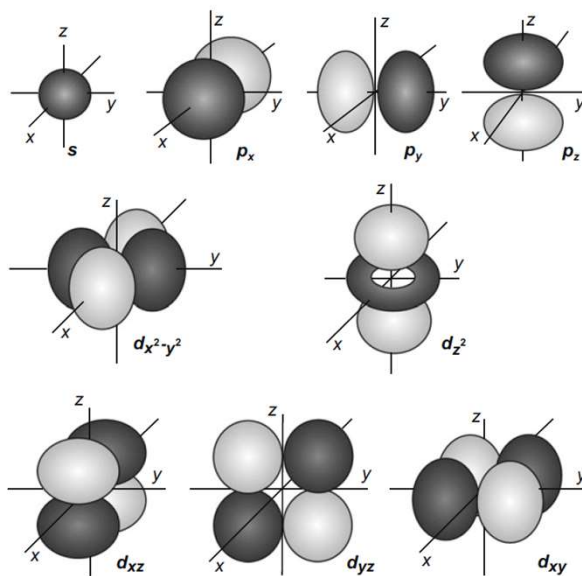
Orbital	n	l	m	$R_{n,l}(r)^a$	$Y_{l,m}(\theta, \phi)^b$
1s	1	0	0	$(Z/a_0)^{3/2} 2 \exp(-\rho)$	$(2\sqrt{\pi})^{-1}$
2s	2	0	0	$\frac{(Z/a_0)^{3/2}}{2\sqrt{2}} (2 - \rho) \exp(-\rho/2)$	$(2\sqrt{\pi})^{-1}$
2p _z	2	1	0	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/4\pi)^{1/2} \cos \theta$
2p ₋	2	1	-1	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/8\pi)^{1/2} \sin \theta \{ \cos \phi - i \sin \phi \}$ $= (3/8\pi)^{1/2} \sin \theta \exp(-i\phi)$
2p ₊	2	1	1	$\frac{(Za_0)^{3/2}}{2\sqrt{6}} \rho \exp(-\rho/2)$	$(3/8\pi)^{1/2} \sin \theta \{ \cos \phi + i \sin \phi \}$ $= (3/8\pi)^{1/2} \sin \theta \exp(i\phi)$

^a $\rho = Zr/a_0$, where Z is the nuclear charge, r is the distance from the nucleus and a_0 is the Bohr radius (0.529 Å)

^b θ and ϕ are the angles with respect to the z and x axes in polar coordinates (Fig. 4.4)

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Atomic orbitals



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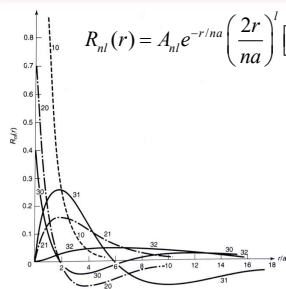
Atomic orbitals

$$n = 1, 2, 3, \dots$$

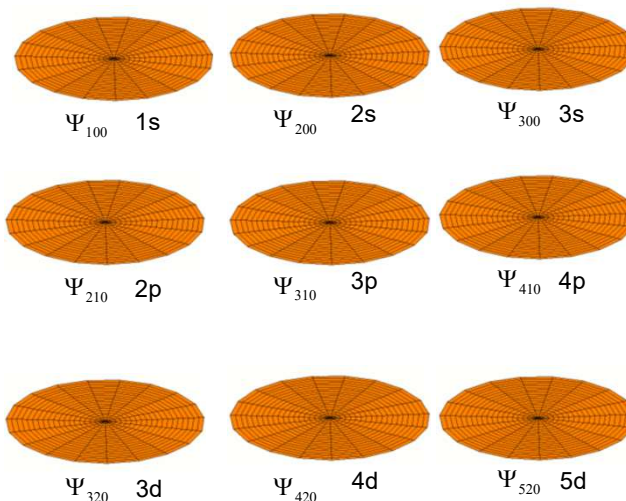
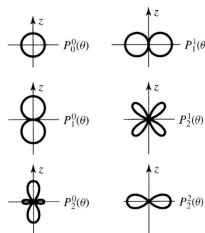
$$l = 0, 1, 2, \dots, n-1$$

$$m = -l, \dots, 0, \dots, +l$$

[standing waves on a circular drum](#)



$$\Theta(\theta) = AP_l^m(\cos \theta)$$



http://en.wikipedia.org/wiki/Atomic_orbital

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